

IFN647 Week 4 Review Questions

Question 1 Which of the following is wrong? and justify your answer.

- (1) The index is inverted because usually we think of words being a part of documents, but if we invert this idea, documents are associated with words.
- (2) In an inverted index that contains only document information, i.e., the features are binary, meaning they are 1 if the document contains a term, 0 otherwise. This inverted index contains enough information to tell if the document contains the exact phrase “tropical fish”.
- (3) Each index term has its own inverted list that holds the relevant data for that term. Each list entry is called a posting, and the part of the posting that refers to a specific document or location is often called a pointer.
- (4) An extent is a contiguous region of a document. We can represent these extents using word positions.

Question 2 (Query processing)

Inverted indexing is an efficient data structure to represent documents for information retrieval, where each index term is associated with an inverted list that contains a list of pairs of document number and count of the term occurrences in that document. The following table is an inverted index for 4 documents and their index terms.

and	1:1				only	2:1			
aquarium	3:1				pigmented	4:1			
are	3:1	4:1			popular	3:1			
around	1:1				refer	2:1			
as	2:1				referred	2:1			
both	1:1				requiring	2:1			
bright	3:1				salt	1:1	4:1		
coloration	3:1	4:1			saltwater	2:1			
derives	4:1				species	1:1			
due	3:1				term	2:1			
environments	1:1				the	1:1	2:1		
fish	1:2	2:3	3:2	4:2	their	3:1			
fishkeepers	2:1				this	4:1			
found	1:1				those	2:1			
fresh	2:1				to	2:2	3:1		
freshwater	1:1	4:1			tropical	1:2	2:2	3:1	
from	4:1				typically	4:1			
generally	4:1				use	2:1			
in	1:1	4:1			water	1:1	2:1	4:1	
include	1:1				while	4:1			
including	1:1				with	2:1			
iridescence	4:1				world	1:1			
marine	2:1								
often	2:1	3:1							

Assume the abstract model of ranking is

$$R(Q, D) = \sum_i g_i(Q) f_i(D)$$

where f_i is the document topical feature function (the value of $f_i(D)$ is term t_i 's counts in document D) and g_i is a query topical feature function ($g_i(Q) = 1$ if term t_i is in query Q ; otherwise, $g_i(Q) = 0$). Let query $Q = \{\text{freshwater, fish, aquarium, tropical}\}$. Calculate each document's ranking score by using Term-at-a-time Algorithm. You are required to write the calculating process (or steps).

Question 3 (Markov chain)

Assume when John is sad today, which isn't very usual: he either goes for a run, gobbles down ice cream or takes a nap next day. The Markov Chain depicted in the following **state diagram** has 3 possible states: "Sleep", "Run", and "Ice Cream".

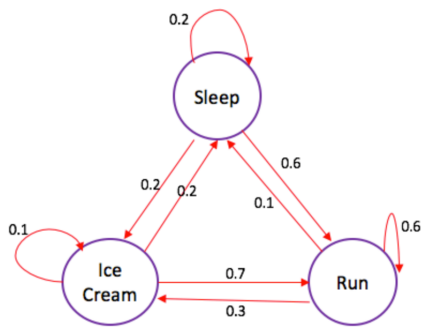


Fig 2. Markov chain example.

The following is the corresponding **transition matrix**

CURRENT STATE	NEXT STATE		
	SLEEP	RUN	ICE CREAM
SLEEP	0.2	0.6	0.2
RUN	0.1	0.6	0.3
ICE CREAM	0.2	0.7	0.1

- (1) According to the above diagram, if John spent sleeping a sad day away, what is the probability of he will likely go for a run next day?
- (2) The transition matrix will be a 3×3 matrix. To simple represent the matrix, we can use the following variables, for example,

SS – Sleep $\rightarrow$ Sleep

SR – Sleep → Run

SI – Sleep → Ice Cream

...

Then the transition matrix for the states is represented as a list of lists in python
[["SS","SR","SI"], ["RS","RR","RI"], ["IS","IR","II"]]

Please write the corresponding transition matrix for probabilities in a list of lists.

Question 4 (Optional) Markov chain cont.

We can represent the transition matrix of Q3.3 as follows:

```
import numpy as np
import random as rm
```

```
# The statespace
states = ["Sleep", "Icecream", "Run"]
# Possible sequences of events
transitionName = [ ["SS", "SR", "SI"], ["RS", "RR", "RI"], ["IS", "IR", "II"] ]
# Probabilities matrix (transition matrix)
transitionMatrix = [[0.2, 0.6, 0.2], [0.1, 0.6, 0.3], [0.2, 0.7, 0.1]]
```

```
if sum(transitionMatrix[0])+sum(transitionMatrix[1])+sum(transitionMatrix[2]) != 3:
    print("Somewhere, something went wrong. Transition matrix, perhaps?")
else: print("All is gonna be okay, you should move on!! ;)")
```

All is gonna be okay, you should move on!! ;)

Design a function to forecast the system state after k days. For example, you may assume that the initial state is "Sleep". You may use “numpy.random.choice” to randomly sample a transition from the set of possible state transitions for each day. Your function prints the results in the same format as the sample output shown below.

Start state: Sleep

Possible states: ['Sleep', 'Run', 'Run']

End state after 2 days: Run

Probability of the possible sequence of states: 0.3

Question 5 (LLM-based Information Extraction)

Given the following XML document:

```
<newsitem itemid="780718" id="root" date="1997-08-06" xml:lang="en">
<title>EU: Official Journal contents - OJ C 237 of August 4, 1997.</title>
<headline>Official Journal contents - OJ C 237 of August 4, 1997.</headline>
<text>
<p>*</p>
<p>Common position (EC) No 28/97 of 17 June 1997 adopted by the Council, acting in
accordance with the procedure referred to in Article 189b of the Treaty establishing
the European Community, with a view to adopting a European Parliament and Council
Directive on the legal protection of designs</p>
<p>Common position (EC) No 29/97 of 17 June 1997 adopted by the Council, acting in
accordance with the procedure referred to in Article 189b of the Treaty establishing
the European Community, with a view to adopting a European Parliament and Council
Directive amending Directives 76/116/EEC, 80/876/EEC, 89/284/EEC and 89/530/EEC on the
approximation of the laws of the Member States relating to fertilizers END OF
DOCUMENT.</p>
</text>
<copyright>(c) Reuters Limited 1997</copyright>
</newsitem>
```

Design an LLM-based information extraction prompt to extract entities “Itemid”, “Organization”, “Date” and “Copyright” from the XML document, and return the result in a valid Python dictionary as follows:

```
{
  "Itemid": 780718,
  "Organization": "Reuters Limited",
  "Location": "European Community",
  "Date": "1997-08-06",
  "Copyright": "(c) Reuters Limited 1997"
}
```